

Group 9 Assignment Report

An Evolutionary Algorithm for the Travelling Salesman Problem

Christos Zampounis Mehmet Cihan Yalcin Péter Dávid Hajdu
Pierre Kasparian Prajval Parthiban

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1 Introduction

The goal of this assignment is to design, implement and experimentally evaluate an evolutionary algorithm for a combinatorial optimisation problem. We chose the Travelling Salesman Problem, whose discrete, permutation-based search space and relevance to routing and logistics make it a natural and challenging testbed for evolutionary computation. This report presents our problem formulation, the evolutionary approach and operators we developed, and an experimental analysis of its solution quality and convergence behaviour.

2 The Travelling Salesman Problem

2.1 Mathematical Formulation

The Travelling Salesman Problem (TSP) is a classical combinatorial optimisation problem that, despite its simple formulation, is difficult to solve: given a set of N cities and the distances between them, the task is to find the shortest closed tour that visits every city exactly once. The problem is NP-hard and no exact algorithm can solve it in polynomial time. Following the formulation of Potvin [1], we model the problem on a complete weighted graph $G = (V, E)$, where $V = \{1, \dots, N\}$ is the set of cities and d_{ij} is the distance between cities i and j . The decision variables are $x_{ij} \in \{0, 1\}$, where $x_{ij} = 1$ if and only if the tour travels directly from city i to city j . The objective is to minimise the total travelled distance,

$$\text{Minimise} \quad \sum_{i=1}^N \sum_{j=1}^N d_{ij} x_{ij}, \quad (1)$$

subject to the degree constraints

$$\sum_{j=1}^N x_{ij} = 1, \quad i = 1, \dots, N, \quad \sum_{i=1}^N x_{ij} = 1, \quad j = 1, \dots, N, \quad (2)$$

which require every city to have exactly one incoming and one outgoing edge, and the subtour-elimination constraints

$$\sum_{i,j \in S} x_{ij} \leq |S| - 1, \quad \forall S \subset V, \quad 2 \leq |S| \leq N - 1, \quad (3)$$

which forbid partial cycles over any proper subset S of the cities. Without constraints (3), the model reduces to the assignment problem and its optimal solution generically consists of several disconnected subtours rather than a single tour. For symmetric instances, such as

the Euclidean TSP (ETSP) considered in this report, the optimal solution of the assignment-problem relaxation typically contains many two-vertex subtours; such instances are therefore better tackled by specialised algorithms that exploit this structure [1].

Because an evolutionary algorithm operates on orderings rather than on binary edge variables, it is convenient to state the equivalent tour-as-permutation formulation. A candidate solution is a permutation $\pi = (\pi_1, \dots, \pi_N)$ of the cities, and its cost is the length of the cycle that visits them in that order,

$$C(\pi) = \sum_{k=1}^{N-1} d_{\pi_k, \pi_{k+1}} + d_{\pi_N, \pi_1}. \quad (4)$$

The evolutionary algorithm therefore seeks a permutation minimising Equation (4); this representation is the one adopted throughout this report.

2.2 Related Evolutionary Approaches

2.2.1 Edge-Swapping Genetic Algorithm (Liu, 2014)

Liu [2] builds a genetic algorithm around a crossover operator called *Edge Swapping* (ES), which keeps the edges shared by the two parent tours and swaps in the cycles where they disagree. The population is initialised not randomly but with a greedy local search using the 2-opt neighbourhood, and on large benchmarks the algorithm reaches best-known solutions. Following Liu, we therefore seed our own population with greedy nearest-neighbour tours refined by 2-opt.

2.2.2 Genetic-based Particle Swarm Optimization (Liao et al., 2012)

Liao et al. [3] propose GPSO, a genetic-based particle swarm optimisation in which each particle is a real-valued vector decoded into a tour by ranking its components. We adopt their geometric initialisation: cities are sorted by their angle around the geographical centre, producing a non-crossing route.

2.2.3 Genetic Algorithm with Order Crossover (Hussain et al., 2017)

Hussain et al. [4] compare crossover operators for the path representation, where a chromosome is the visiting order and classical one-point or uniform crossovers cannot be used as they duplicate cities. They review partially mapped (PMX), order (OX) and cycle crossover, and propose a modified CX2, but no operator dominated on TSPLIB. We adopt OX: a segment of one parent is copied into the child and the remaining cities are filled in their order of appearance in the other parent, always yielding a valid tour.

2.3 Motivation of the Developed Approach

The evolutionary approach we develop follows the genetic algorithm presented in Section 9.2 of Mitchell [5], adapted to the permutation representation of the TSP; its pseudo-code is given in Section 3. Within this template, the design choices left open concern the initialization of the population and the crossover operator, which we align with the literature reviewed in Section 2.4. We initialize the population from informed tours rather than at random, combining greedy nearest-neighbour tours refined by 2-opt [2] with geometric, non-crossing tours [3]; for recombination we adopt the order crossover (OX) [4]. The remaining operators are left unchanged, so that the effect of the two adapted components can be isolated in Section 4.

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3 Algorithm

Each individual is represented as a permutation of all cities, where the order of the cities defines the tour and the final city is connected back to the first. The initial population consists of p randomly generated permutations. The cost of an individual is given by Equation (4). Since the TSP is a minimisation problem, this cost is converted into a maximisation-based fitness function defined as

$$F(\pi) = \frac{1}{C(\pi)}. \quad (5)$$

Consequently, shorter tours receive higher fitness values and are favoured during selection.

In each generation, a fraction $(1 - r)$ of the population is selected and carried over unchanged. The remaining fraction r is replaced by offspring generated from selected parent pairs. Tournament selection is used to select parents. For each parent a small number of individuals are sampled randomly from the population, and the individual with the highest fitness is selected. This process is repeated to obtain the second parent. Tournament selection was chosen because it is simple to implement and provides selection pressure towards shorter tours while maintaining population diversity.

Order Crossover (OX) is used to generate offspring because it preserves valid permutations. A segment from one parent is copied to the offspring, while the remaining cities are inserted according to their order in the second parent, skipping already used cities. Two offspring are generated for each parent pair by swapping the parent roles.

After crossover, swap mutation is applied to a fraction m of the new population. Two randomly selected positions in an individual are exchanged, which introduces variation while preserving a valid permutation. The process is repeated for G generations, and the best tour found during the run is returned.

Algorithm 1 Genetic Algorithm for the Travelling Salesman Problem

Require: cities, population size p , replacement rate r , mutation rate m , number of generations G

Ensure: best tour found

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 $P \leftarrow p$  random permutations of the cities
evaluate the fitness of each individual in  $P$ 
for  $g = 1$  to  $G$  do
     $P_s \leftarrow$  the best  $(1 - r)p$  individuals from  $P$ 
    for  $i = 1$  to  $(rp)/2$  do
        select parents  $h_1, h_2$  from  $P$  using tournament selection
         $o_1 \leftarrow \text{ORDERCROSSOVER}(h_1, h_2)$ 
         $o_2 \leftarrow \text{ORDERCROSSOVER}(h_2, h_1)$ 
        add  $o_1$  and  $o_2$  to  $P_s$ 
    end for
    apply SWAPMUTATION to a fraction  $m$  of  $P_s$ 
     $P \leftarrow P_s$ 
    evaluate the fitness of each individual in  $P$ 
    update the best tour found
end for
return best tour

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4 Experimental part

5 Results and Analysis

6 Conclusions

References

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